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Why Normalization Exists

Before normalization existed, databases often stored the same information repeatedly.

This led to:

- Wasted storage
- Incorrect updates
- Missing information
- Inconsistent data
- Difficult maintenance

Normalization provides a systematic process for organizing data so that:

- Each fact is stored only once.
- Relationships between data are maintained correctly.
- Redundancy is minimized.
- Data remains consistent and reliable.

What is Redundancy?

Definition

Redundancy is the unnecessary duplication of the same data in multiple places within a database.

Example

Student	Department	HOD
Aadya	CSE	Dr. Sharma
arpit	CSE	Dr. Sharma
aadz	CSE	Dr. Sharma
Aadya rastogi	ECE	Dr. Mehta

Why is it a problem?

Suppose Dr. Sharma is replaced by Dr. Singh.
You now need to update every CSE student's record.

Redundancy causes:

- Wasted Storage
- More Updates
- Slower Maintenance
- Inconsistent Information

Data Anomalies

A. Update Anomaly

Definition
Updating one fact requires updating multiple rows

B. Insertion Anomaly

Definition
You cannot insert certain information without inserting unrelated information.

Example
Suppose a new department is created:

- AI Department
- HOD = Dr. Kapoor

But no students have joined yet.
If the table only stores department details along with students, you cannot record the new department.
This is an Insertion Anomaly.

C. Deletion Anomaly

Definition
Deleting one piece of data unintentionally deletes other important information.

Example

Student	Department	HOD
Aadya	AI	Dr. Kapoor

If Aadya graduates and her row is deleted:

```
DELETE FROM Student;
```

The only information about the AI department is also lost.
This is a Deletion Anomaly.

Database Consistency

Definition

Consistency means the database always represents the same real-world truth. Normalization stores each fact once, making it easier to keep the database consistent.

Data Integrity

Definition

Data integrity is the accuracy, correctness, and reliability of the data stored in a database.

Types of Integrity

- **Entity Integrity**
  Every row has a unique primary key.
- **Referential Integrity**
  Foreign keys reference valid rows.
- **Domain Integrity**
  Values follow valid formats.
- **User-defined Integrity**
  Custom business rules.

Normalization supports integrity by reducing duplication and making constraints easier to enforce.

Storage Efficiency

Suppose there are:

- 100,000 students
- 50 departments

Without normalization:
Each student stores:

- Department Name

- HOD
- Building
- Office Number

This information is repeated 100,000 times.

Goals of Normalization

Normalization aims to:

- Reduce redundancy.
- Eliminate anomalies.
- Improve consistency.
- Improve integrity.
- Simplify updates.
- Save storage.
- Make maintenance easier.

Is redundancy always bad?

Answer

No. In OLTP systems, redundancy is generally minimized through normalization. However, in data warehouses, reporting systems, analytics platforms, and caching layers, controlled redundancy (called denormalization) is intentionally introduced to improve read performance. We'll study this in Chapter 19 — Denormalization.